



WeCloudData

Agent Evaluation

Introduction





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Why Agent Evaluation Is Different Agent Evaluation



Why Do Agents Need Evaluation?

Agent Evaluation

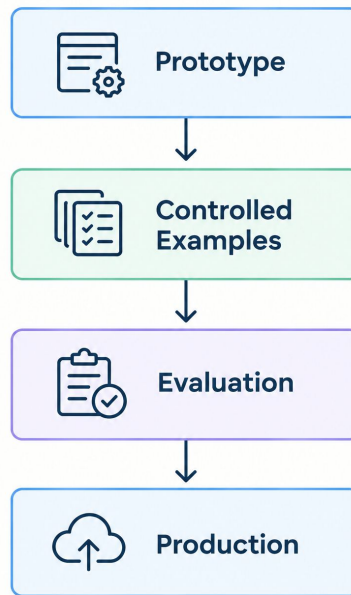
Evaluation tells us whether an agent works reliably.

Production introduces:

- Unpredictable inputs
- Real-world constraints
- Safety risks
- Latency and cost
- Changing tools and environments

Evaluation helps verify:

- Reliability
- Safety
- Consistency
- Robustness

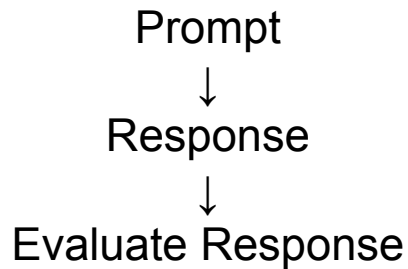




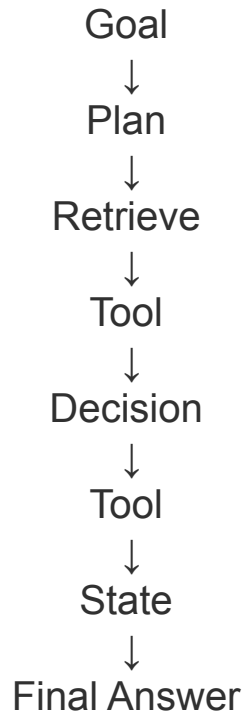
Why Agent Evaluation Is Different

Agent Evaluation

Traditional LLM



Agent



A correct final answer does not guarantee a correct execution.





Outcome vs Process Evaluation

Agent Evaluation

Agent Evaluation

Final Outcome:

Did it succeed?

+

Process

How did it succeed?

Evaluate:

- Final outcome
- Observable decisions
- Tool calls
- State transitions
- Execution path
- Final result

Outcome evaluation is necessary but insufficient.

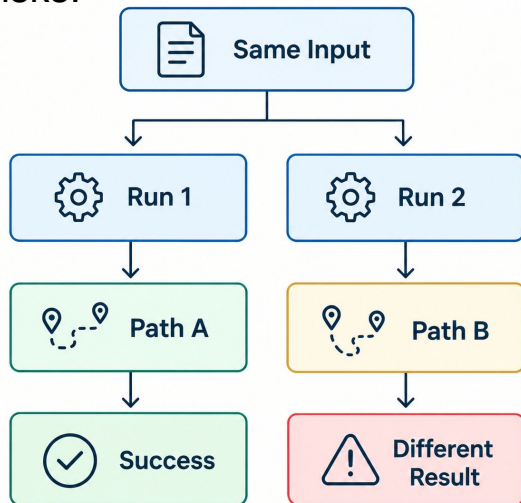




Agents Are Stochastic

Agent Evaluation

Summarize this report and email the key risks.



Why paths can differ:

- Probabilistic model behavior
- Dynamic context
- Tool responses
- Execution decisions

One successful run does not establish reliability.



Success $\neq$ Correct Behavior

Agent Evaluation



Examples:

- Email claimed as sent
- Database claimed as updated
- File claimed as created
- Deployment claimed as completed

An agent can appear successful without actually completing the task.



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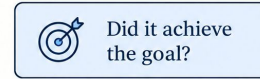
What Does Success Mean? Agent Evaluation



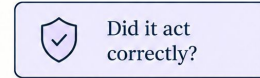
What Does Agent Success Mean?

Agent Evaluation

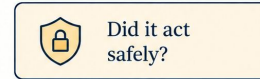
When do we say an agent succeeded?



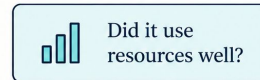
+



+



+



Agent success is more than producing a correct answer.



Five Practical Evaluation Dimensions

Agent Evaluation

Accuracy

correct result

Reliability

works
consistently

Safety

avoids harmful
actions

Efficiency

avoids
unnecessary
steps

Cost

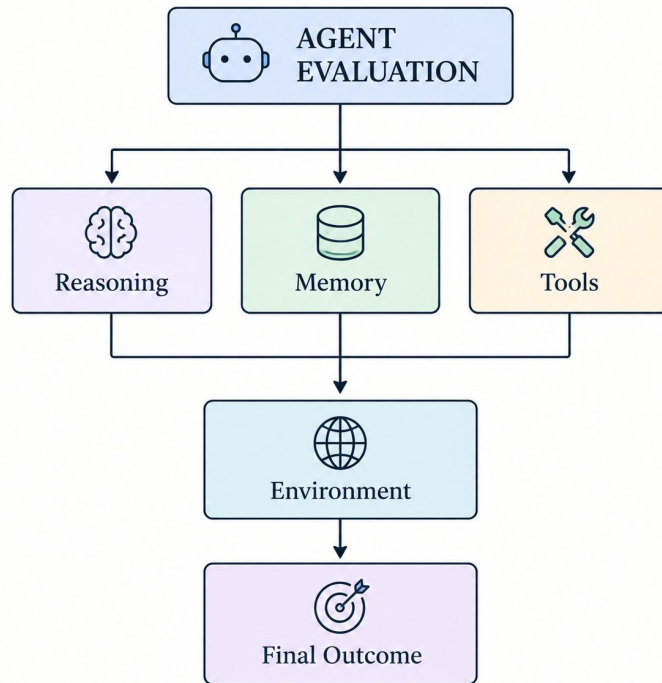
reasonable
compute/token
usage





Four Layers of Agent Evaluation

Agent Evaluation



An agent can fail at any layer.



Goal Alignment and Safety

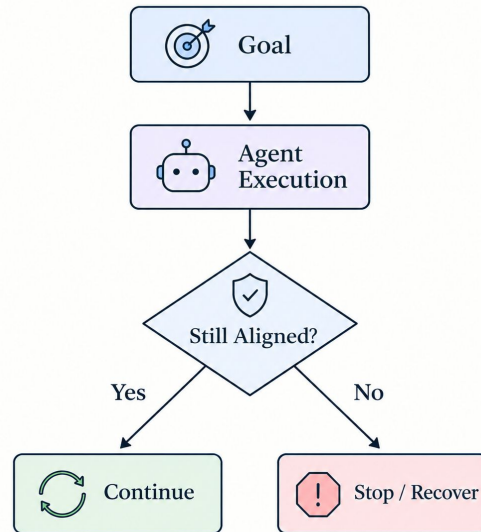
Agent Evaluation

Goal Alignment

- Did the agent stay focused on the user's goal?
- Did it follow user intent?
- Did it avoid unnecessary goal drift?

Safety

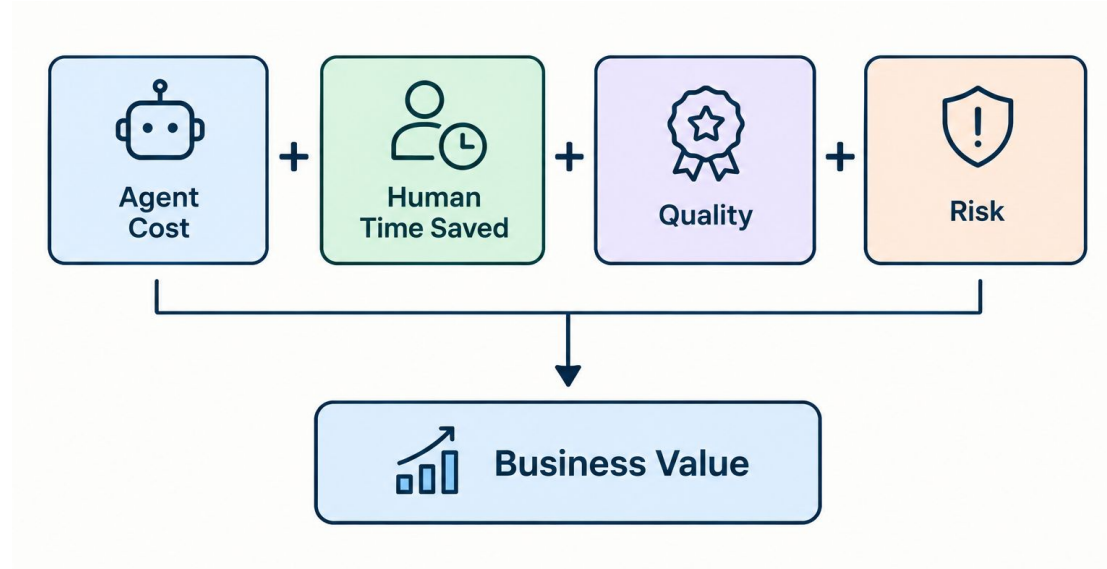
- Did it follow policies?
- Did it avoid unsafe actions?
- Did it protect sensitive information?





Business Value and Efficiency

Agent Evaluation



A successful agent should create meaningful value at an acceptable cost and risk.



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Evaluation Datasets & Benchmarks

Agent Evaluation



What Is an Agent Evaluation Dataset?

Agent Evaluation

Evaluation Dataset



Tasks + Expected
Outcomes



Agent



Execution



Metrics

Traditional NLP
datasets often
focus on:

- Text input
- Text output

**Agent evaluation
datasets also test:**

- Multi-step tasks
- Tool usage
- Planning
- Memory/state
- Environment interaction

Agent datasets evaluate what the agent does, not only what it says.





Benchmark Landscape

Agent Evaluation

Benchmark	Main Focus
SWE-bench	Software engineering agents
GAIA	General assistant tasks
MCP-Bench	Tool-connected agents
AndroidWorld	Mobile agents
Synthetic datasets	Controlled evaluation

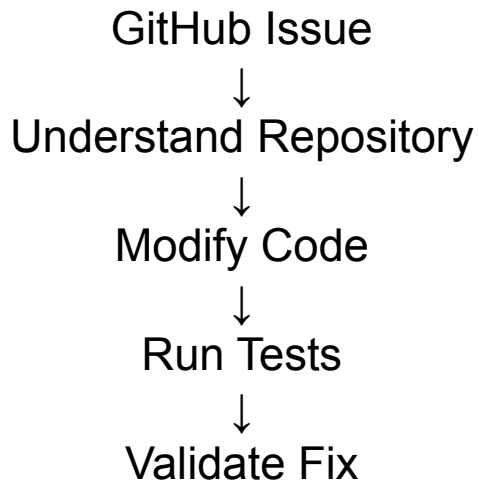
A benchmark measures the capability it was designed to test.





SWE-bench

Agent Evaluation



Measures agent ability to:

- Understand coding tasks
- Work with repositories
- Modify existing code
- Run tests
- Validate solutions





GAIA

Agent Evaluation

General-purpose agents must combine reasoning, tools, and information sources.

Question



Search



Read



Calculate



Reason



Answer

Examples:

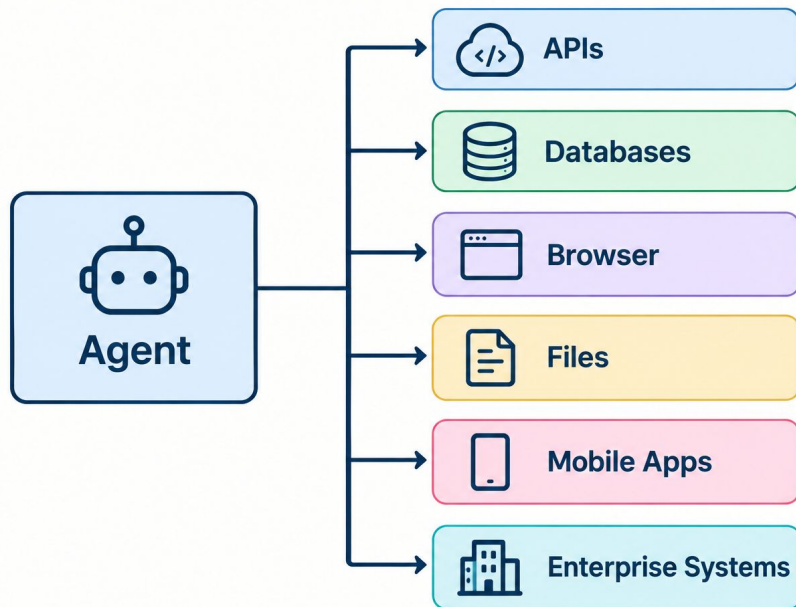
- Research a company
- Analyze a spreadsheet
- Plan travel
- Compare information sources





Environment

Agent Evaluation



The environment becomes part of the evaluation.



Synthetic Evaluation

Agent Evaluation

Real Data	Synthetic Data
More realistic	More controlled
Expensive	Lower cost
Privacy concerns	Easier to scale
Harder to collect	Easy to generate
Real edge cases	May miss edge cases





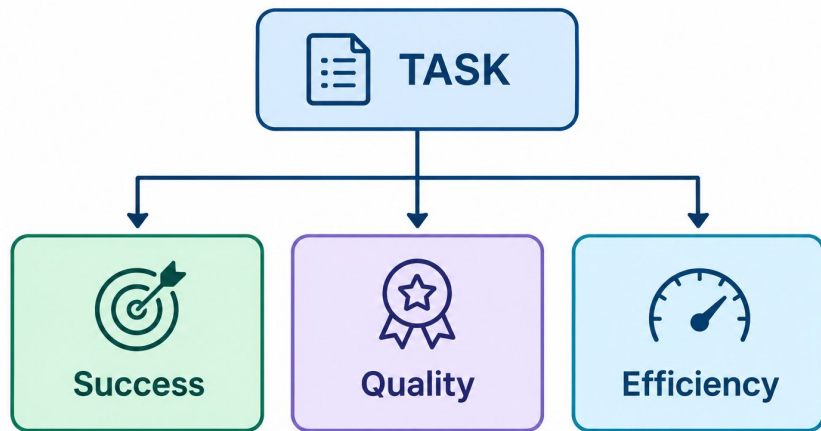
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How Do We Measure an Agent? Agent Evaluation



Task-Level Success Measurement

Agent Evaluation



Ask:

- Did the task succeed?
- Was the process appropriate?
- Were resources used efficiently?
- Did tools and memory work correctly?

An agent can appear successful without actually completing the task.





Trace-Level vs Span-Level Metrics

Agent Evaluation

Trace-level

- Entire workflow
- Task completion
- End-to-end success

Span-level

- Individual actions
- Retrieval
- Tool calls
- API usage

Trace tells us whether the task succeeded. Span-level analysis helps explain why.





Trace-Level Metrics

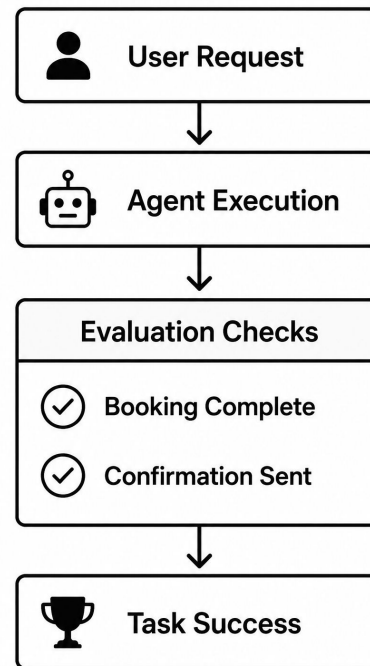
Agent Evaluation

Evaluate the entire workflow from start to finish.

Measures:

- task completion
- verified task outcome
- end-to-end faithfulness
- overall success rate

Trace-level metrics identify whether a task succeeded.





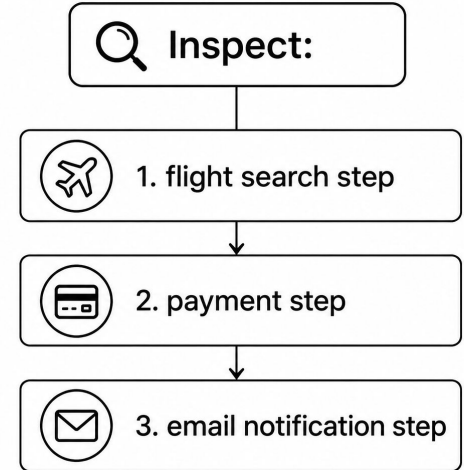
Span-Level Metrics

Agent Evaluation

Evaluate individual actions within the workflow.

Measures:

- retrieval quality
- tool-call correctness
- observable decisions and plan updates
- API usage



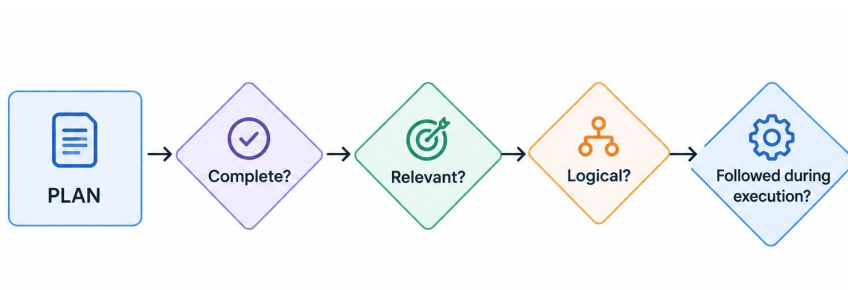
Span-level metrics explain why a task succeeded or failed.



Reasoning and Planning Metrics

Agent Evaluation

Instead of trying to score hidden reasoning, evaluate observable plans and decisions.



Metrics:

- Plan completeness
- Relevance
- Logical structure
- Plan adherence
- Useful execution decisions

An agent can appear successful without actually completing the task.



Tool and Action Metrics

Agent Evaluation

Tool Selection



Argument Correctness



Execution Validity



Progress

Measure:

- Correct tool selection
- Correct parameters
- Valid execution
- Redundant calls
- Unnecessary actions
- Infinite loops

A successful tool call is not necessarily a correct tool call.



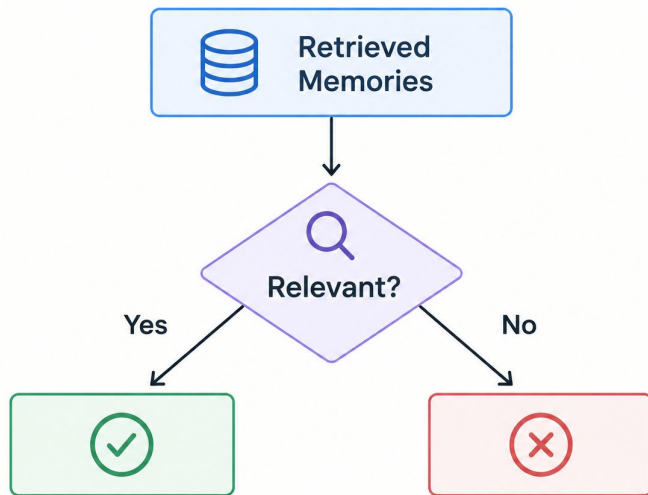


Memory Evaluation

Agent Evaluation

Retrieval quality

Did the agent retrieve the right information?





Memory Evaluation

Agent Evaluation

Memory usefulness

Did the agent use the retrieved information correctly?

Then:

- **Precision** → how much retrieved information is relevant
- **Recall** → how much relevant information was retrieved





Memory Evaluation

Agent Evaluation

Single-hop:

Customer Profile

Multi-hop:

Customer Profile
+
Transaction History
+
Support Tickets



Step Efficiency

Agent Evaluation

Task



Actual Path



Compare with Useful /
Necessary Steps

Measure:

- Number of steps
- Unnecessary actions
- Redundant tool calls
- Excessive retries
- Distance from an efficient execution path

Avoid claiming there is always one objectively “optimal” path.





pass@k vs pass^k

Agent Evaluation

Example:

Run 1 ✓
Run 2 ✗
Run 3 ✓
Run 4 ✓

pass@4

Did **at least one** run
succeed?

Yes ✓

pass⁴

Did **every** run
succeed?

No ✗

pass@k measures the chance of finding a successful run.

pass^k highlights consistency across runs.





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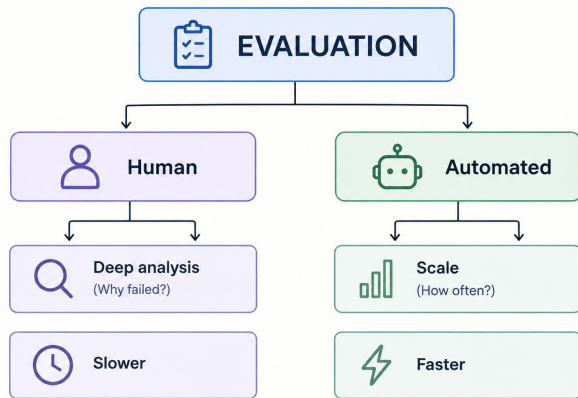
Evaluation in Practice

Agent Evaluation



Automated vs Manual Evaluation

Agent Evaluation



Human Evaluation

- Deep analysis
- Failure investigation
- Complex judgments

Automated Evaluation

- Large-scale testing
- Repeated evaluation
- Continuous monitoring

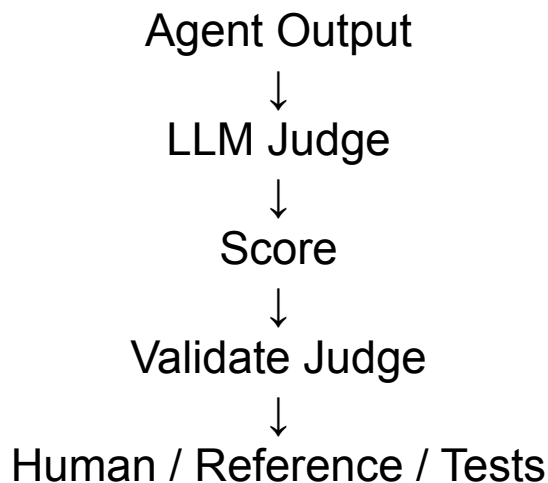
Manual evaluation helps explain failures. Automated evaluation helps measure them at scale.





LLM-as-a-Judge

Agent Evaluation



Benefits

- Fast
- Scalable
- Can apply a standardized rubric
- Reduces human effort

Challenges

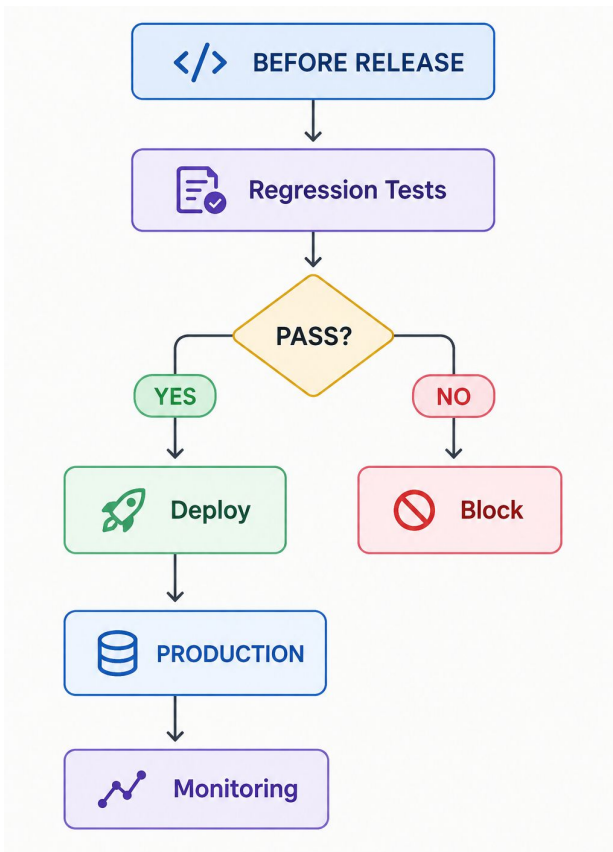
- Model bias
- Style/verbosity bias
- Score inconsistency
- Judge may disagree with humans or references

Never assume the judge is ground truth.



Regression Gates and Production Monitoring

Agent Evaluation





Regression Gates and Production Monitoring

Agent Evaluation

Regression evaluation

- Task success
- Safety
- Tool accuracy
- Latency
- Benchmark performance

Production monitoring

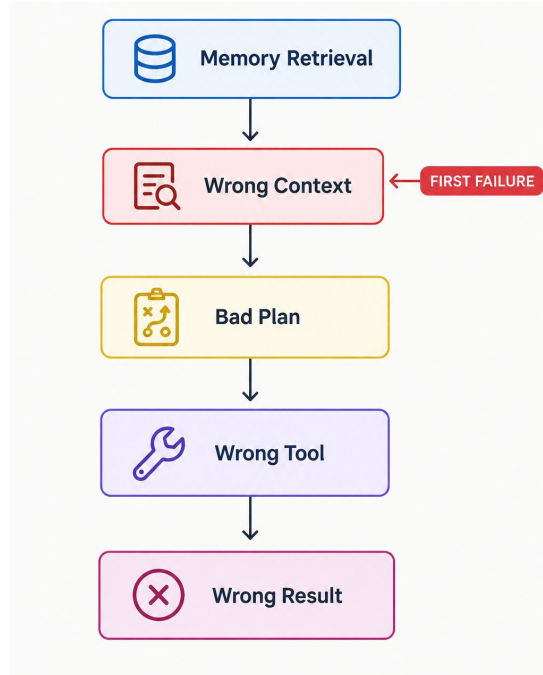
- New failures
- Performance changes
- Cost
- Latency
- Safety events

Regression testing protects releases.
Production monitoring protects live systems.



Find the First Meaningful Failure

Agent Evaluation



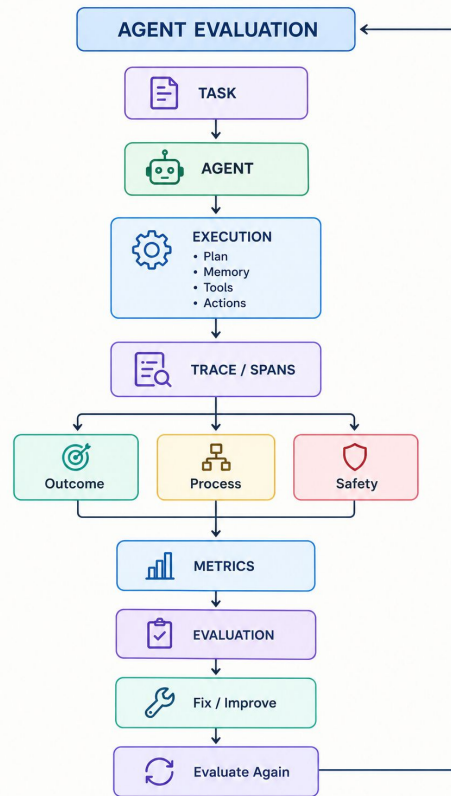
Don't only measure where the task failed. Find where the failure began.



Complete Agent Evaluation Loop

Agent Evaluation

Evaluate → Diagnose → Improve →
Re-evaluate → Release or Block





Summary

- Agents need evaluation because real-world tasks are complex.
- Agent evaluation looks at both outcomes and execution.
- Success means being accurate, reliable, safe, and efficient.
- Different benchmarks test different agent capabilities.
- Task-level metrics measure overall success.
- Trace-level metrics show the complete execution.
- Span-level metrics help identify individual problems.
- Tools, memory, planning, and safety should all be evaluated.
- One successful run does not prove reliability.
- Evaluation helps us find problems and improve the agent.



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Thank You

We'd love to hear from you. If you have any questions,
feel free to reach out to our team anytime!